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Luiz Augusto Facury de Souza

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Research Interests

Self-supervised learning, JEPAs, world models, latent-space planning, ECG representation learning.

Education

Universidade Federal de Minas Gerais (UFMG), MSc in Computer Science — Jan 2025 – Present

- Advisor: Prof. Wagner Meira Jr.; thesis on self-supervised, JEPA-based world models and latent-space planning.
- Current work: online JEPA-based methods for Atari, using SIGReg regularization.
- Coursework: Deep Learning, Computer Vision, Design and Analysis of Algorithms.

Télécom Paris (IP Paris), Diplôme d'Ingénieur (MSc in Engineering) — Sept 2022 – Sept 2024

- Double degree with UFMG in Data Science, Image Processing and AI; taught in English and French.
- Coursework: Deep Learning, Statistics, Optimization, Databases, Programming Paradigms.
- End-of-studies project on self-supervised dense segmentation graded 19/20; led to a publication at MIDL 2024.

Universidade Federal de Minas Gerais (UFMG), BS in Electrical Engineering — Aug 2018 – Dec 2024

- Specialization in ML and Computer Engineering; coursework in Operating Systems and Electronics.

Publications

stable-worldmodel: A Platform for Reproducible World Modeling Research and Evaluation 2026

Lucas Maes*, Quentin Le Lidec*, **Luiz Facury**, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, Randall Balestriero

*Neural Information Processing Systems (NeurIPS) 2026 – **Poster**. arXiv:2605.21800 (*equal contribution)*

Learning Navigable World Models via Latent Energy Shaping 2026

Luiz Facury, Jose Geraldo Fernandes, Pedro Dutenhefner, Gisele L. Pappa, Wagner Meira Jr.

2nd Workshop on World Models @ ICLR 2026

Beyond Patient Invariance: Learning Cardiac Dynamics via Action-Conditioned JEPAs 2026

Jose Geraldo Fernandes, **Luiz Facury**, Pedro Dutenhefner, Gisele L. Pappa, Wagner Meira Jr.

2nd Workshop on World Models @ ICLR 2026

Clinically Interpretable Zero-Shot ECG Classification via Multimodal Learning and Expert-Aligned Descriptors 2025

Luiz Facury de Souza, Jose Geraldo Fernandes, Pedro Robles Dutenhefner, Turi Andrade Rezende, Gisele L. Pappa, Gabriela Miana Paixão, Antonio Luiz Pinho Ribeiro, Wagner Meira Jr.

*Computing in Cardiology (CinC) 2025. Workshop version, "Bridging the Gap in Clinical AI: Zero-Shot Multimodal ECG Analysis with Textual Explanations", at LXAI @ ICML 2025 – **Oral Presentation***

Transferring Clinical Knowledge into ECGs Representation: A Self-Supervised Approach for Interpretable, Unimodal-at-Inference Diagnosis 2025

Jose Geraldo Fernandes, **Luiz Facury de Souza**, Pedro Dutenhefner, Gisele L. Pappa, Wagner Meira Jr.

Time Series for Health (TS4H) Workshop @ NeurIPS 2025

Curved Spaces, Enhanced Diagnosis: Hyperbolic Neural Networks for Multi-Label ECG Classification 2025

Pedro Dutenhefner, Diogo Tuler, Turi Rezende, Jose Fernandes, **Luiz Facury**, Luísa Porfírio, Yan Aquino, Arthur Buzelin, Pedro Bento, Gabriela Paixão, Gisele L. Pappa, Antonio Ribeiro, Wagner Meira Jr.

Computing in Cardiology (CinC) 2025

Dense Self-Supervised Learning for Medical Image Segmentation 2024

Maxime Seince, Loïc Le Folgoc, **Luiz Augusto Facury de Souza**, Elsa Angelini

Medical Imaging with Deep Learning (MIDL) 2024

Multiple-Input Multiple-Output Randomized Fuzzy Cognitive Map Method for High-Dimensional Time Series Forecasting 2024

Omid Orang, Hugo Vinicius Bitencourt, **Luiz Augusto Facury de Souza**, Patrícia de Oliveira Lucas, Petrônio C. L. Silva, Frederico Gadelha Guimarães

IEEE Transactions on Fuzzy Systems

Combining Embeddings and Fuzzy Time Series for High-Dimensional Time Series Forecasting in Internet of Energy Applications 2023

Hugo Vinicius Bitencourt, **Luiz Augusto Facury de Souza**, Matheus Cascalho dos Santos, Rodrigo Silva, Petrônio C. L. Silva, Frederico Gadelha Guimarães

Energy (Elsevier)

An Embedding-Based Non-Stationary Fuzzy Time Series Method for Multiple Output High-Dimensional Multivariate Time Series Forecasting in IoT Applications 2023

Hugo Vinicius Bitencourt, Omid Orang, **Luiz Augusto Facury de Souza**, Petrônio C. L. Silva, Frederico Gadelha Guimarães

Neural Computing and Applications (Springer)

Experience

Researcher, SPEED Lab, UFMG – Belo Horizonte, Brazil Aug 2024 – Present

- Self-supervised and multimodal models for 12-lead ECG diagnosis, with Hospital das Clínicas (UFMG).
- Led a zero-shot, text-aligned ECG classifier that matched a supervised baseline (accuracy 0.968 vs. 0.973) while explaining each diagnosis with clinical criteria (CinC 2025; ICML 2025 workshop oral).
- Extended this line of work to world models: action-conditioned JEPAs for cardiac dynamics and latent energy shaping for navigation (two papers at the ICLR 2026 World Models workshop).

Researcher I, Petrobras – Belo Horizonte, Brazil Jan 2025 – Present

- Built a framework to locate and segment regions of interest in geological data for oil exploration (2 papers under review).

Research Intern, Philips – Paris, France Feb 2024 – Aug 2024

- Developed a self-supervised foundation model for ECGs; pre-training on unannotated recordings improved the model in the company's diagnostic pipeline and reduced reliance on costly expert labels.
- Submitted two patent applications on applying the foundation model to automated clinical diagnosis.

Summer Research Intern, Sorbonne Université – Paris, France June 2023 – July 2023

- Built a vision system (ResNet and ViT encoders) to track a single-celled protozoan and keep it in focus.

Research Intern, Computational Intelligence Laboratory, UFMG – Belo Horizonte, Brazil July 2021 – Aug 2022

- Deep learning for text-to-speech, speech-to-text and face (lip) synchronization.
- Developed a Portuguese speech synthesis and transcription model despite scarce annotated data.

Software Engineering Intern, Infosistemas – Belo Horizonte, Brazil Jan 2021 – July 2021

- Developed, debugged and optimized the company's backend API with Node.js and SQL.

Open-Source Software

- **stable-worldmodel** (co-author, NeurIPS 2026): world-model research platform, 2.2k+ GitHub stars.
- **stable-pretraining** and **stable-datasets** (contributor): with Randall Balestriero's group, Brown University.

Service, Skills and Languages

Reviewer: World Models Workshop @ ICLR 2026; Physical Understanding for Decision-Making @ NeurIPS 2026

Machine learning: SSL (contrastive, JEPA), multimodal learning, world models, computer vision, time series

Programming and tools: Python, PyTorch, Git, LaTeX, SQL, Node.js

Languages: Portuguese (native), English (C1), French (C1)

Volunteering: Mathematics mentor, Eductus (University of São Paulo), 2020, during COVID-19 school closures.